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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

ME1006-1 – MANUFACTURING PROCESSES

(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following statements is true?
A) Casting is the replica of the object to be cast B) Pattern is the replica of the casting object
C) Casting and the pattern are the same thing D) Molten material is cast into the casting cavity
2. What is the main advantage of sand casting?
A) High accuracy B) Low cost
C) High surface finish D) Ability to produce complex shapes
3. What is the purpose of the gating system in sand casting?
A) To melt the metal B) To transfer the metal into the cavity
C) To create the mold cavity D) To remove the casting from the mold
4. What is the purpose of the riser in sand casting?
A) To melt the metal B) To pour the metal into the mold
C) To create the mold cavity D) To feed the casting during solidification
5. Which of the following is a principal binder used in carbon dioxide moulding?
A) Carbon B) Carbon Dioxide
C) Silica sand D) Sodium silicate
6. Which of the following casting process uses a polystyrene pattern?
A) Investment casting B) Ceramic-shell casting
C) Evaporative-pattern casting D) Squeeze casting
7. Why is flux added to the charge (parent metal to be melted) during the melting of a metal?
A) To increase carbon content B) To increase refractoriness
C) To add specific oxides D) To remove specific oxides
8. There is no direct contact of flame and metal in which of the following types of furnace?
A) Cupola B) Crucible
C) Electric arc D) Induction
9. In Which Type of Extrusion Process Extrusion pressure is transmitted through the fluid to the billet:
A) Hydrostatic extrusion B) Impact extrusion
C) Backward or Indirect Extrusion D) Extrusion forging
10. Which type of rollers is mainly used in Sheet Rolling
A) Three-high rolling mill B) Two-high rolling mill
C) Four-high rolling mill D) Tandem rolling mill
11. In which of the following forging methods special die is not used?
A) Closed die forging B) Smith forging
C) Machine forging D) Drop forging
12. Cutting and forming operations can be performed in a single operation in a
A) Simple die B) Progressive die
C) Compound die D) Combination die

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13. Which is the disadvantage of a ceramic material?
 - A) Hard
 - B) Brittle
 - C) Chemically stable
 - D) Refractory
14. In jiggering process, initial shape is obtained using a
 - A) Punch
 - B) Bat former
 - C) Spinneret
 - D) Combination die
15. The initial tube in blown film extrusion is obtained by
 - A) Injection molding machine
 - B) Rotomolder
 - C) Blower
 - D) Extruder
16. Polymer spinning process is used to make
 - A) Sheets
 - B) Discrete items
 - C) Fibers
 - D) Continuous product
17. What are the two materials joined in explosive welding?
 - A) Metals only
 - B) Plastics only
 - C) Metals and plastics
 - D) Glass and metals
18. What is thermit welding?
 - A) A welding process that uses an electric arc to melt the metals
 - B) A welding process that uses a gas flame to heat the metals
 - C) A welding process that uses a chemical reaction to produce heat and melt the metals
 - D) A welding process that uses ultrasonic vibrations to bond the metals
19. What is the typical joint configuration used in ultrasonic welding?
 - A) Butt joint
 - B) Lap joint
 - C) T-joint
 - D) V-joint
20. Which type of metal is commonly used as a filler metal in brazing?
 - A) Copper
 - B) Silver
 - C) Zinc
 - D) Aluminum

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain briefly the steps involved in the casting process.	8	L*2	1	1
	b) Explain true centrifugal casting process with a sketch.	8	L2	2	1
2.	a) Explain the various terms used in the casting process with a sketch.	8	2	1	1
	b) Write a note on any four casting defects.	8	2	2	1
3.	a) Write a note about loose piece pattern and match plate pattern.	8	2	1	1
	b) Write a note on electric arc furnace.	8	2	2	1
Unit – II					
4.	a) Compare hot and cold working of metals.	8	4	3	1
	b) Compare metals with non-metals.	8	4	4	1
5.	a) Explain the working of a compound die with a sketch.	8	2	3	1
	b) Explain float process of glass manufacturing.	8	2	4	1
6.	a) Explain rod and wire drawing process.	8	2	3	1
	b) Explain the extrusion process of plastics.	8	2	4	1
Unit – III					
7.	a) Explain the process of TIG welding with a sketch.	8	2	5	1
	b) Write a note on the different types of oxy-acetylene flames.	8	2	5	1
8.	a) Explain the process of MIG welding with a sketch.	8	2	5	1
	b) Explain the explosive welding process with a sketch.	8	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
